

Project Progress

Nowcasting Inflation via Daily Web Scraping Price

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1. Objective

- Build an Automated Data Infrastructure and Monitoring System:** Set up an automated data pipeline and database to collect, store, and continuously monitor prices.
- Calculate CPI for Inflation Nowcasting Using Machine Learning Model:** Apply an economic formula to compute daily price relatives, and use regularized Ridge regression modeling to nowcast inflation ahead of the official monthly release.
- Build a Decision-Support Dashboard:** Develop an interactive dashboard that presents daily CPI results and inflation for easy monitoring.

2. Architecture

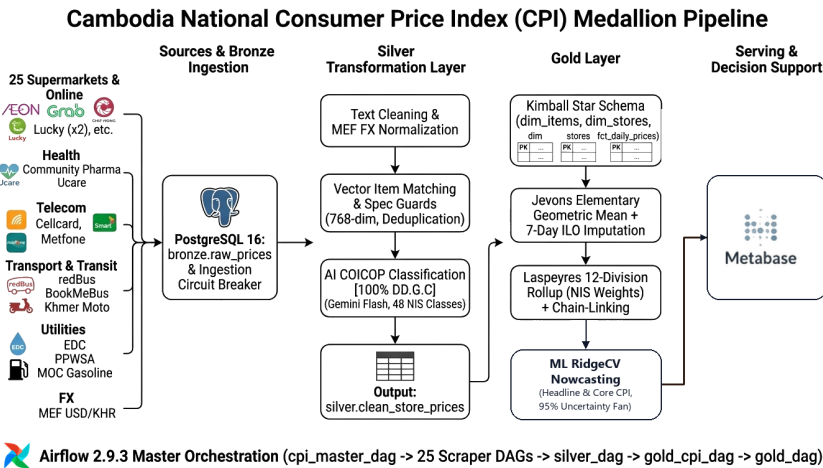


Figure 1: End-to-End System Architecture: Web Scraping, Medallion Pipeline & Dashboard.

Technology Stack

Component / Pipeline Layer	Technology Stack
Data Ingestion & Scraping	Python
Database and Storage	PostgreSQL 16
Orchestration	Apache Airflow
Data Transformation	dbt Core
AI Product Classifications	Gemini AI
Dashboard and Monitoring	Metabase
System Container	Docker

3. Data Collection

Data Source	Description
AEON Online Cambodia	Supermarket
DeliShop Asia Phnom Penh	Supermarket
GrabMart (Lucky Supermarket, Chip Mong)	Supermarket
L192 Online Marketplace	Footwear & Homeware
Community Pharmacy & Ucare Pharmacy	Healthcare & Pharmaceutical Goods
Khmer24 & Realestate.com.kh	Urban Residential Apartment, House Rentals
redBus & BookMeBus Transit	Bus & Travel Transport
Cellcard, Smart Axiata, Metfone	Mobile Plan and Wi-Fi
Tela Khmer	Gasoline
dataEF API	Exchange Rate
Sokha, Hyatt, Bayon BKK	Restaurant and Hotel

4. Methodology

Stage 1: Web Scraping Layer

Automated Daily Collection: Every morning at 08:00 AM, the system automatically collects prices across 11 key stores and services in Cambodia.

How We Collect Prices:

- **Direct Store APIs:** Connects directly to the backend of online supermarkets (AEON, DeliShop) and apps (L192, BookMeBus) to download product lists and prices quickly without loading heavy web pages.
- **Browser Disguise (curl_cffi):** Sends fast automated requests that look like a real Google Chrome browser so store security systems (like Cloudflare) do not block our computer.
- **Invisible Browser (Playwright):** If a website blocks direct requests with security checks (like 403 Forbidden), the system automatically opens an invisible browser to load the page and read the prices like a real user.
- **Public Pages & Feeds:** Reads apartment rental listings from Khmer24, fuel prices from the Tela Khmer public channel (translating Khmer numbers 0–9), and official daily exchange rates from the Ministry of Economy and Finance (MEF).
- **Safety & Backup Rules:** Pauses 1 second between requests so we do not slow down store websites, and sends an alert if 0 products are found so bad data is never saved.

Stage 2: Bronze Layer (Raw Lake Ingestion)

Database Table: bronze.raw_prices

- **What it does:** Acts as an immutable, append-only raw data repository preserving the exact raw data as fetched from the web.
- **Stored Fields:** store_id, source_name, item_description_raw, price, currency, source_url, scraped_at, batch_id.

Stage 3: Silver Layer (Standardization, AI Classification & Quality Control)

Database Table: silver.standardized_prices

The Silver Layer has 3 main jobs:

[Raw Scrape] → 1. Clean Data → 2. Match Product → 3. Classify Basket → [Clean Silver Item]

1. Data Cleaning & Standardization

- **Clean product names and text:** Remove HTML, special characters, promotional words (SALE, HOT DEAL, 50% OFF), unnecessary text, and convert text to lowercase.
- **Convert Khmer numerals → Arabic numerals:** Convert Khmer digits (e.g., ៥០០ → 500) for consistent processing.
- **Convert USD → KHR:** Convert dual currencies using the official MEF daily exchange rate.
- **Standardize weight, volume, and package units:** Convert package sizes into standard metric units (e.g., 500g to 0.5kg, 1500ml to 1.5L).
- **Calculate unit prices (KHR/kg, KHR/L, etc.):** Compute unit price ($P_{unit} = \text{Price in KHR} / \text{Normalized Metric}$) to fairly compare different product sizes.

2. Product Matching (Find the Same Item)

Connects identical products across different stores (e.g., AEON vs. DeliShop) using a 5-step waterfall:

- **Level 1 — Barcode (GTIN/EAN):** Exact barcode match.
- **Level 2 — Store SKU:** Match with past days’ catalog.
- **Level 3 — Clean Name & Size:** Exact text match with size protection (prevents 330ml matching 1.5L).
- **Level 4 — AI Vector (pgvector):** Matches English and Khmer equivalents ($\geq 95\%$ similarity).
- **Level 5 — Fuzzy Spelling:** Catches small typos.

3. Classify into Official Baskets (UN COICOP)

- Uses Google Gemini AI in batches to assign the official 4-digit government code (e.g., 01.1.1.1 for Rice/Bread).
- **Strict Guardrails prevent AI mistakes:**
 - Pet Food → Goes to Pets (09.3.1), never human food.
 - Shampoo / Soap → Goes to Personal Care (12.1.3), never food.
 - Beer / Wine → Goes to Alcohol (02.1), isolated from soft drinks.

The Gold Layer in Brief: From Clean Products to Official CPI

The Gold Layer applies official international standards (ILO / UN guidelines) to turn thousands of clean daily prices into national economic indicators in 3 main steps:

[Clean Silver Items] → 1. Jevons Index → 2. Laspeyres Aggregation → 3. Core CPI & Marts

1. Elementary Index Calculation (Jevons Formula)

- **Jevons Geometric Mean:** Instead of a simple average (which gets distorted by expensive luxury items), the system uses the geometric mean:

$$I_{\text{rice},t} = \left(\frac{P_{1,t}}{P_{1,0}} \times \frac{P_{2,t}}{P_{2,0}} \times \dots \times \frac{P_{n,t}}{P_{n,0}} \right)^{\frac{1}{n}}$$

- **Missing Price Imputation:** If an item is temporarily out-of-stock, a 7-day carry-forward rule fills in the price so the index doesn’t jump wildly.

2. Official National Aggregation (Laspeyres Formula)

Cambodians spend more money on food than on furniture. To reflect real life, the 12 COICOP divisions are combined using official government survey expenditure weights (CSES 2020):

$$\text{Headline CPI} = \sum \left(\text{Weight} \times \text{Division Index} \right)$$

- **Food & Drinks (01):** 44.775% (Highest weight in Cambodia)
- **Housing & Electricity (04):** 17.062%
- **Transport & Fuel (07):** 12.203%
- **All other 9 Divisions:** Remaining 25.960% (Clothing, Health, Telecom, etc.)

3. Central Bank Metric: Core CPI

For central bankers at the National Bank of Cambodia (NBC), temporary global oil spikes or seasonal vegetable floods distort the real picture.

- **Refined Core CPI:** The system calculates a second official index that removes Food (01) and Energy (04):

$$\text{Core CPI} = \frac{\sum_{\text{ex-Food, ex-Energy}} w_j \cdot I_j}{\sum w_j}$$

This shows the underlying, long-term price stability of the economy.

Stage 4: Nowcasting

- **Methodology (Two-Tier Hybrid Approach):** Rather than using a black-box AI model, we use a hybrid economic approach:
 - *Tier 1 (Official Aggregation):* We use fixed CSES expenditure weights across all 12 COICOP divisions following official NIS/ILO Laspeyres standards. We calculate both Headline CPI and Core CPI (excluding volatile food and energy).
 - *Tier 2 (Forward Econometric Projection):* For past observed days, we compute the realized daily average from scraped prices. For the unobserved remaining days of the month, we nowcast daily drift rates using regularized Ridge Regression (RidgeCV) with Bayesian prior shrinkage.
 - *Blending:* We combine realized days and projected days using a time-weighted formula that smoothly converges onto actual data as the month ends.
- **Cambodia-Specific Market Features:**
 - *5 Core Baskets:* We model the 5 high-velocity categories (~81.6% of Cambodia's basket): Food (44.8%), Housing/Utilities (17.1%), Transport (12.2%), Restaurants (3.1%), and Alcohol (1.6%).
 - *Freight Spillover:* Fuel price momentum (Transport) is selectively fed into Food and Restaurant catering to model logistics cost pass-through.
 - *Holiday Handling:* We built an exponential decay kernel for Khmer holidays (Khmer New Year, Pchum Ben, Water Festival) so temporary ceremonial spikes are not misread as permanent inflation.
 - *FX Pass-Through:* Ingests daily USD/KHR commercial exchange rates.

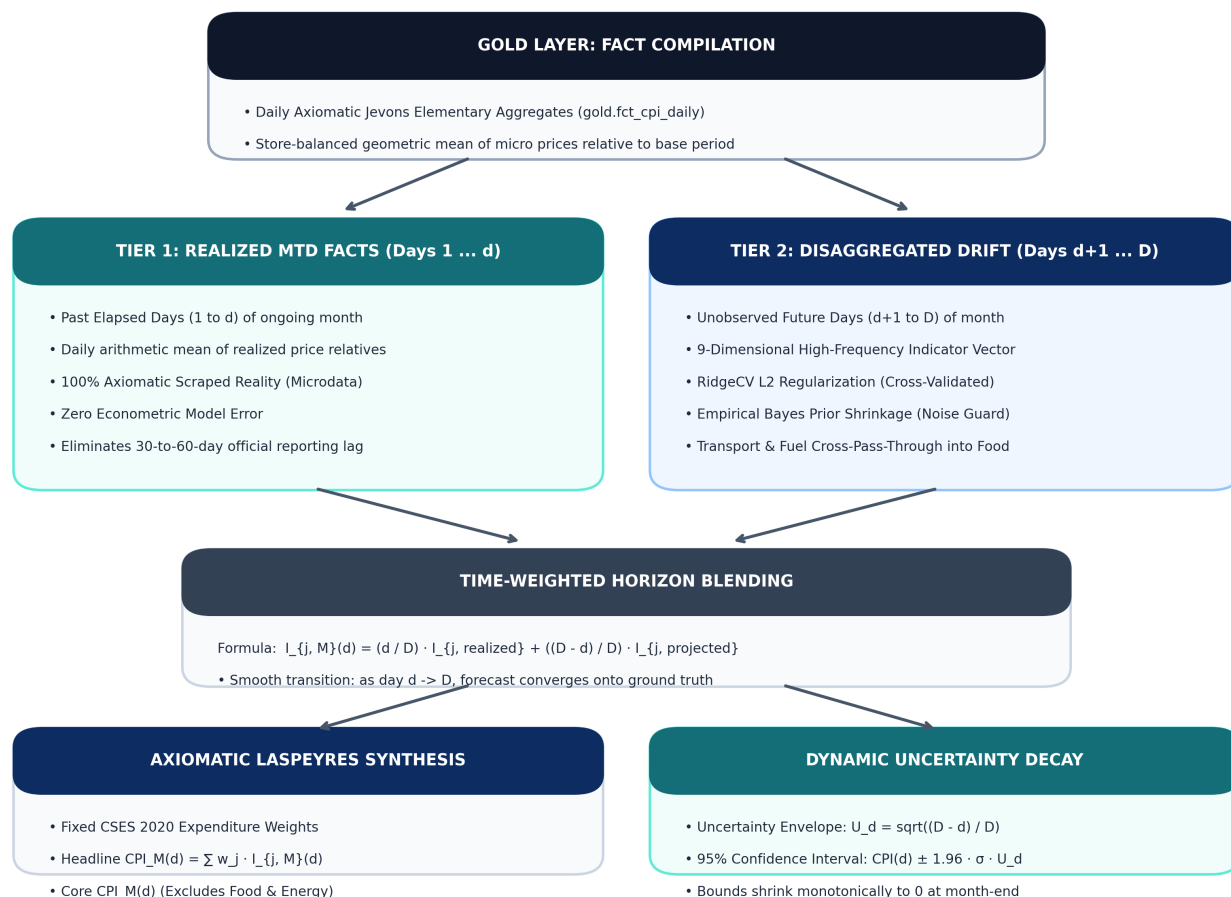


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Figure 2: Architecture Flowchart of the Two-Tier Inflation Nowcasting Engine

Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence & Uncertainty Decay

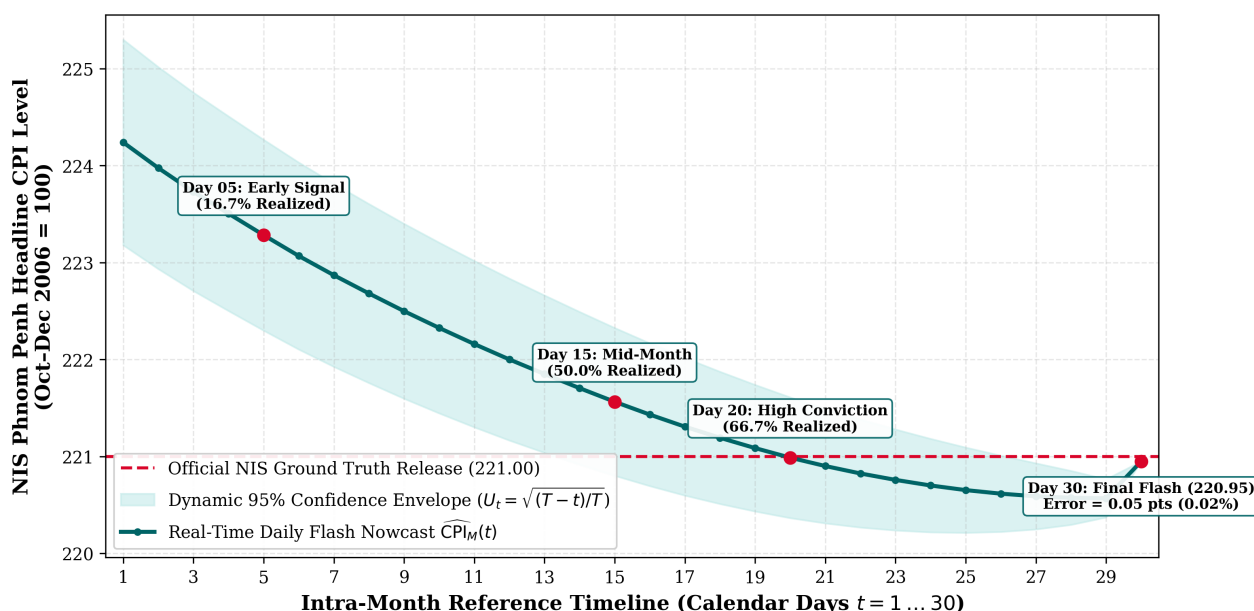


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Figure 3: Intra-Month Nowcasting Convergence with Monotonically Decaying Error Bounds

5. Decision-Support Metabase Dashboards & Monitoring

To provide actionable macroeconomic intelligence and continuous pipeline observability, the system implements three dedicated production dashboards in Metabase (deployed via Docker containers):

Dashboard 1: Macro CPI & Inflation Analytics

Shows daily Headline CPI and Core CPI (which removes volatile food and fuel prices), monthly inflation rates, and price changes across all 12 consumption categories compared against official government reports.

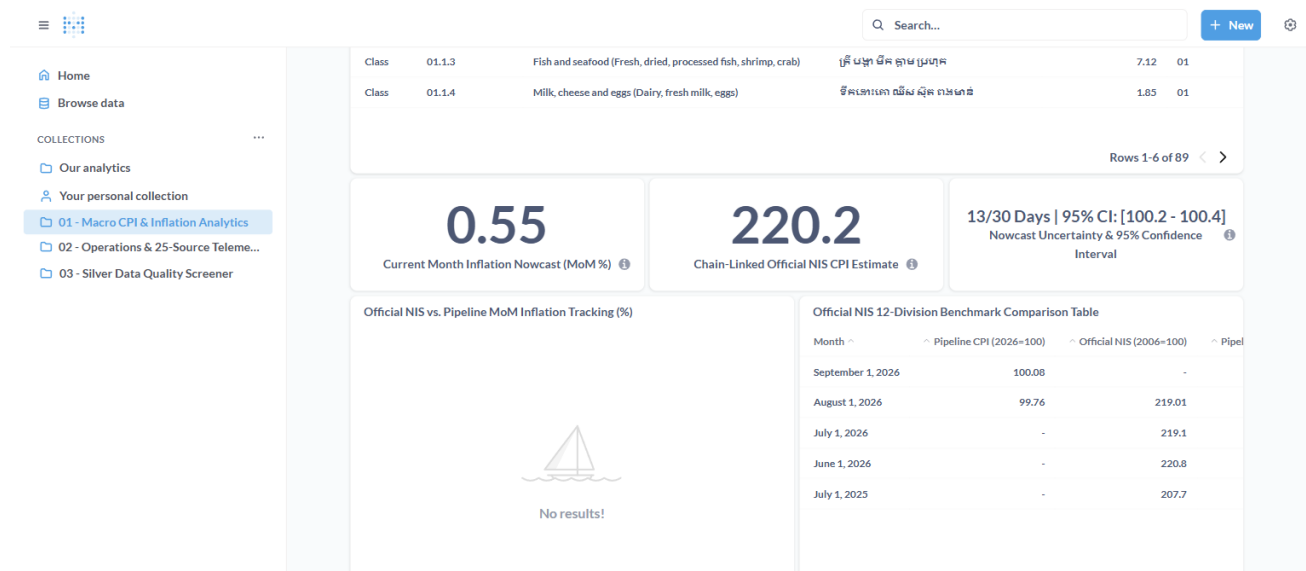


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Figure 4: Metabase Dashboard 1 — Real-Time Macro CPI & Inflation Analytics

Dashboard 2: Operations & Data Source Telemetry

Tracks the daily health of our scrapers across all 11 sources. It shows how many product prices were collected each day, scraping speed, and website connection status.

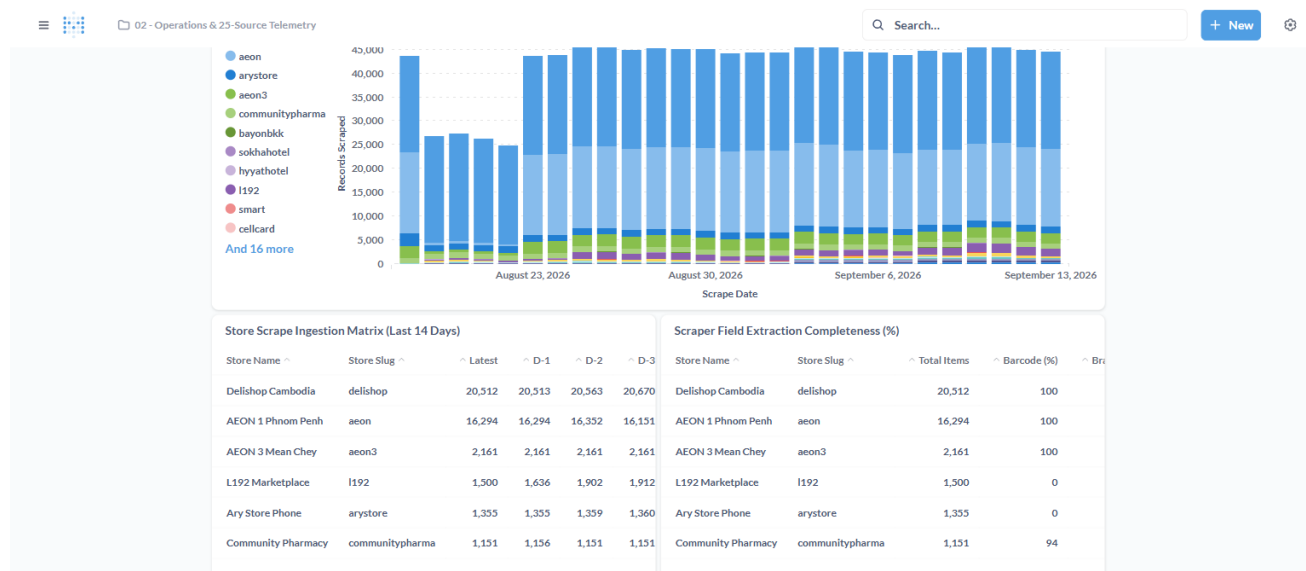


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Figure 5: Metabase Dashboard 2 — Data Lake Operations & Scraping Telemetry

Dashboard 3: Silver Layer Data Quality Screener

Checks the cleanliness of our data every day. It tracks product size standardization (converting items to 1kg or 1L), USD to Khmer Riel currency conversion, product matching across stores, and AI classification accuracy.

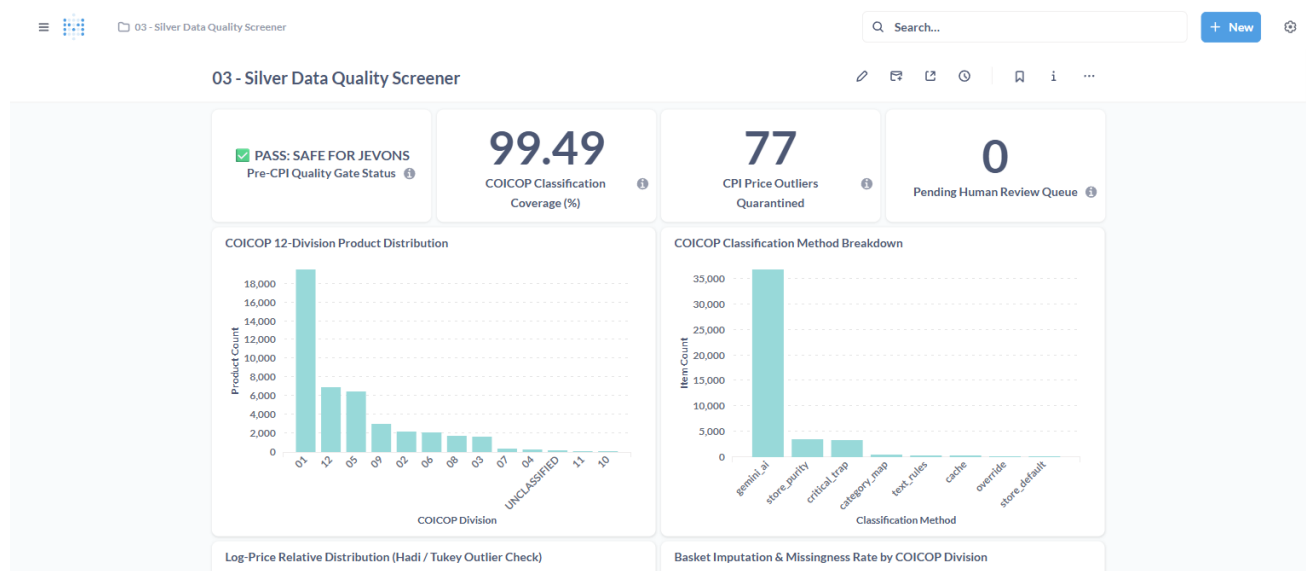


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Figure 6: Metabase Dashboard 3 — Silver Layer Data Quality & Classification Screener